

PABLO ALEJANDRO REYES GRANADOS

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Education

Universidad Externado de Colombia 2021–2025

B.A. in Economics

- Highest GPA in the Economics cohort.
- Concentration in Quantitative Methods and Artificial Intelligence applied to Economics.

Universidad Externado de Colombia 2024–Present

M.Sc. in Business Intelligence

- Focus on advanced analytics, machine learning, statistical modeling, and data management.

Universidad Externado de Colombia 2023–Present

Undergraduate Track in Data Science

- Formal training in programming, data science, machine learning, statistical modeling, and computational methods.

Massachusetts Institute of Technology (MITx) 2025–Present

MicroMasters Program in Statistics and Data Science

- Rigorous training in probability, statistics, data analysis, and machine learning.

National University of Colombia Aug 2025–Dec 2025

Diploma in Advanced Machine Learning and Data Science

- Advanced applied training in machine learning, including NLP, 3D computer vision, MLOps, and end-to-end ML projects.

Research Experience

Professional Research Assistant | Central Bank of Colombia – Board of Directors 2025–Present

Bogotá, Colombia · Hybrid

- Conduct empirical research using causal inference, survival analysis, and high-dimensional financial datasets to study market behavior, credit risk, and institutional dynamics.
- Develop survival and hazard models to analyze default selection, delinquency transitions, and risk dynamics across large-scale household credit portfolios.
- Build reproducible Python pipelines for data cleaning, feature engineering, statistical modeling, and analysis of large-scale financial and regulatory datasets.

Research Intern | Central Bank of Colombia – Board of Directors Jul 2025–Dec 2025

Bogotá, Colombia · Hybrid

- Conducted applied research on uniform-price auction microstructure, focusing on bidding behavior, liquidity provision, and price formation.
- Constructed and validated analytical datasets for econometric analysis of auctions, dealer behavior, and primary-market dynamics.
- Applied causal and microeconomic methods to study strategic behavior in high-frequency institutional trading environments.

Junior Researcher | NeuroMinds Research Group, Universidad Externado 2025–Present

Bogotá, Colombia · In person

- Conduct research in computer vision, generative modeling, and representation learning for socially rele-

vant identification tasks, in collaboration with Colombia's Missing Persons Search Unit.

- Develop deep learning pipelines for facial aging, biometric synthesis, and image-based representation learning under limited-data and domain-shift settings.
- Explore diffusion-based and transformer-based methods for robust visual representation learning, with emphasis on controllable generation and human-centered AI applications.

Research Assistant | Universidad Externado de Colombia

Jan 2025–Dec 2025

Bogotá, Colombia · Remote

- Developed quantitative policy analysis using Bayesian econometric methods to assess fiscal sustainability, macroeconomic risk, and policy transmission.
- Designed a graph-based framework for input-output matrices to study sectoral interdependence, shock propagation, and diffusion dynamics across the Colombian economy.
- Built reproducible computational workflows for economic modeling, simulation, and network-based analysis of complex economic systems.

Conference Presentations

"Under the Shadow of the Fed: Colombian Macroeconomic Vulnerability to U.S. Interest Rate Shocks." 2025 Bayes Plurinational Conference [Poster] [Replication]

Teaching Experience

Teaching Assistant | Universidad Externado de Colombia

2023–2024

Bogotá, Colombia · In person

Convex Optimization

- Directed tutorials and designed practical computational exercises on optimization, dynamic programming, and multivariable calculus using Python and SciLab.

Courses and Specializations

DeepLearning.AI

May 2025–Aug 2025

Machine Learning & Deep Learning Specializations

- Training in neural networks, optimization, convolutional networks, sequence models, attention mechanisms, Transformers, and practical deep learning workflows.

Relevant Coursework

Probability, Statistics, Optimization, Machine Learning, Deep Learning, Sequence Models, Attention Mechanisms, Transformers, Bayesian Econometrics, Causal Inference.

Technical Skills

Languages and tools: Python, R, Stata, LaTeX, Git/GitHub, Linux.

Methodologies: Deep Learning, Representation Learning, Computer Vision, Generative Modeling, Causal Inference, Bayesian Econometrics, Survival Analysis, Graph Theory.

Frameworks: PyTorch, scikit-learn, OpenCV, pandas, NumPy, Polars, statsmodels.